Scale-rich network metamaterials grow by local rules on a recursive cell division

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Scale-rich (SR) metamaterials are planar ligament networks grown by nucleating a segment at a random point and angle and extending it until it meets the existing structure, with a thickness that decays with the time of birth. In the dual graph, where ligaments are nodes and junctions are links, the degree distribution has a k^{-3} tail. Here we show that, at zero thickness, the SR construction is Cowan's iterated division of cells with area-weighted selection, and that its dual graph grows by a face-local rule: a new node joins the two nodes that carry two sides of one face. Cowan's uniform split fractions make the cell genealogy a random binary search tree. The degree of an interior ligament equals its number of face incidences, so linear preferential attachment on incidences would give the Barabási–Albert law $p_k = 12/[k(k+1)(k+2)]$. The geometric model does not grow that way. A ligament is hit in proportion to its length, which is fixed at birth. Its degree is close to a Poisson variable with mean $4\ell_i(\sqrt{T}-\sqrt{i})/c_L$: the variance around this mean equals the mean within the errors, the oldest ligaments receive 2% more hits than this mean-field rate, and early and late hits are uncorrelated. Length acts as a fitness that the cell hierarchy assigns at birth, $\ell_i \propto i^{-1/2}$, and the k^{-3} tail follows from it. Using integral geometry we compute the probability that a new ligament cuts off a corner of its cell ($2\ln 2/\pi$ for a square), which sets the triangle closure and the clustering spectrum $C(k) \simeq 2[p_0 + p(k-2)]/[k(k-1)]$. A purely topological face-splitting rule reproduces $P(k)$, $C(k)$ and $k_{nn}(k)$, but through reinforcement and with longer distances. Square-closing splits raise the shortest-path multiplicity above its value at fixed degree sequence. Thickness, jamming, angle sets, alignment and density gradients leave $\gamma = 3$, whereas coherent orientation fields, which make the hit rate grow faster than length, give $\gamma \simeq 2.4$–2.5 up to $T = 10^6$. In three dimensions plates are hit in proportion to their birth area, and $\gamma = 5/2$, whereas preferential attachment on incidences would give 2.44; the mechanism tests again select the fitness picture. At fixed decay exponent, the reported per-ligament energy law cannot distinguish degree from length or age.

I. INTRODUCTION

Both *et al.* [1] introduced scale-rich (SR) mechanical metamaterials, two-dimensional ligament networks in which length, thickness and connectivity span orders of magnitude. The construction has two parameters. At step $t = 1, 2, \ldots, T$ a point is drawn uniformly in the unit square together with a uniform angle, and a ligament of thickness $\lambda_t = \lambda_0 t^{-\alpha}$ is grown both ways from it until each end meets an existing ligament or the boundary. Every junction is a T. Following the road-network convention [2, 3], the authors represent each ligament as a node and each junction as a link. They report power-law distributions of thickness, length and degree, with $P(k) \sim k^{-3}$ attributed to a spatial preferential attachment, because longer ligaments offer more opportunities to be hit. They also report a per-ligament elastic energy law $U_i \sim k_i^{3/2}\lambda_i^{1/2} f(\theta_i)$ and a density scaling of the total energy controlled by α [1].

Two questions are natural from the point of view of network growth. Which growth class does the SR dual graph belong to, and which of its properties are fixed by that class rather than by the design parameters (α, λ_0)? Local growth rules, in which a new node attaches to an existing node and to some of its neighbours, generate preferential attachment without global information, together with a hierarchical clustering spectrum $C(k) \sim 1/k$ and degree correlations [4–7]. Growth with node-specific fitness generates scale-free degrees in a different way, through attractiveness fixed at birth rather than through reinforcement [8]. The geometric construction is itself not new: with zero thickness it is Cowan's iterated cell division [9], a member of the family of random tessellations that also contains the STIT model [10, 11] and related to hierarchical crack patterns formed by successive domain divisions [12, 13]. We show that the SR dual grows by a face-local rule, as local-growth models do, but that its degree law is of the fitness type: the fitness is the ligament length, assigned at birth by the cell hierarchy. Both mechanisms give $\gamma = 3$ in two dimensions; they are told apart by the fluctuations of the degree at fixed length and age. Keeping both references in view, the recursive division and the local graph rule, turns most of the reported network statistics into exact or asymptotically exact statements.

The paper is organised as follows. Section II maps the SR construction onto Cowan's division and derives the statistics of the cell hierarchy. In Sec. III we write the dual growth as a face-local rule, contrast preferential attachment with length as fitness, and test which one the geometry realises. Section IV computes the corner-cut probability by integral geometry and derives the clustering spectrum. Section V asks which properties survive when geometry is removed, and Sec. VI turns to dis-

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tances and shortest-path multiplicity. Section VII identifies which rules move the degree exponent, Sec. VIII treats plates in three dimensions, and Sec. IX discusses what this means for the reported structure–property relations.

II. THE SR CONSTRUCTION IS A RECURSIVE CELL DIVISION

We first set $\lambda_0 = 0$ and return to thickness at the end of this section. The current ligaments partition the square into convex polygonal cells. A uniform point falls in cell K with probability equal to its area A_K. Because K is convex and bounded by ligaments, the new ligament grown from that point stops on the boundary of K: it is the chord of K through a uniform point of K at a uniform angle. The SR model is therefore the iterated division of cells with *area-weighted selection* and the *chord through a uniform point* division rule, model (a) of Fig. 2 in Cowan [9]. Cowan rescales the window after each division, by a factor that keeps the mean cell size fixed, so that a stationary limit exists; the rescaling is a similarity and does not change the topology. STIT tessellations differ in both rules: cells are selected by perimeter and cut by isotropic uniform random lines [10, 11]. Gilbert's model [14], used as a reference architecture in Ref. [1], grows all segments simultaneously and is a different process. Hierarchical crack patterns in drying layers also divide domains one at a time and form T-junctions, but the division is deterministic or nearly so, set by the domain shape and the layer thickness [12, 13]. Throughout, the boundary of the square is not a node of the dual graph; 0.55% of ligament ends lie on it.

Corner numbers. A junction placed on the side of a neighbouring cell is a straight angle there, so the corner number n of a cell is fixed at birth. A chord ending on two sides gives each daughter the two new corners at its ends, so $n_1+n_2 = n+4$. Starting from the square, after t steps there are $t+1$ cells and $4+4t$ corners, so the mean corner number is exactly $\langle n\rangle = 4$. Measured number fractions for $n = 3, \dots, 7$ are $0.301, 0.452, 0.200, 0.042, 0.005$. A Markov-chain treatment of the side number under division, as done for proliferating epithelia [15], would give these fractions once the corner-cut probabilities of Sec. IV are known by side number; the rules differ from the epithelial case because T-junctions do not add corners to the neighbours.

Uniform splitting and the binary search tree. Let K be a convex body, x a uniform point in K, and θ any direction independent of x. For fixed θ, the offset s of the line through x along the normal direction has density $w(s)/A_K$, where $w(s)$ is the chord length at offset s. The area on one side of the line is $\int_{-\infty}^{s} w(s')ds'$, so the area fraction is the cumulative distribution function of s evaluated at s. It is therefore uniform on $(0, 1)$, whatever the shape of K and the law of θ. This uniformity and the resulting exponential law of rescaled areas are Cowan's results [9]. They have a compact restatement. Label each cell by the interval of $[0, 1]$ that it would occupy if areas were laid on a line. With area-weighted selection, the sequence of splits is exactly that of a binary search tree (BST) built from uniform keys, the familiar random scission of an interval: the key falls in an interval with probability equal to its length and splits it at the key [16]. The cell areas after T steps are therefore the $T + 1$ spacings of T uniform points, $P(TA > x) \to e^{-x}$ and $\sum_K A_K^2 = 2/(T + 2)$, and the genealogy of cells inherits the BST statistics: the depth of the split at step t grows as $2\ln t$, with variance $2\ln t$, and areas never multiscale. Our simulations give a depth slope 1.983, a variance slope 1.980 and $\sum A^2 = 9.998 \times 10^{-5}$ at $T = 2 \times 10^4$ against 9.999×10^{-5} [Fig. 2(a,b)]. The multiscaling of Krapivsky and Ben-Naim [17] arises with rectangular cuts and lives in the lengths, not in the areas. Axis-aligned chords reproduce their width exponents $\gamma(a) = (a - \sqrt{a^2+4})/2$ within 2%, while with isotropic angles the perimeter moments scale as $t^{1-a/2}$, without multiscaling.

Lengths and the age of the host. Cells at time t have linear size $t^{-1/2}$, and the mean length of the ligament born at t is $\langle\ell_t\rangle \simeq c_\ell t^{-1/2}$ with $c_\ell = 1.16$. The total length grows as $L(t) \simeq c_L\sqrt{t}$ with $c_L = 2c_\ell = 2.33$. The first ligament has $\langle\ell_1\rangle = 0.945$, the ℓ_0 of Ref. [1]. Section III shows that an end of ligament t lands on a given ligament with probability proportional to its length. Replacing that length by its mean, a mean-field step, the host's birth index h satisfies $P(h) \simeq \ell_h/L(t) \simeq \frac{1}{2}(ht)^{-1/2}$. Then $P(h \le xt) = \sqrt{x}$, that is, asymptotically,

$$h/t \;\to\; U^2, \qquad U \sim \text{Uniform}(0, 1). \tag{1}$$

Equivalently, $\ln(t/h)$ is exponential with mean 2 [Fig. 2(c)]. Measured values of $\langle(h/t)^\alpha\rangle$ are 0.667, 0.498 and 0.330 for $\alpha = 1/4$, $1/2$ and 1, against $1/(1 + 2\alpha) =$ 0.667, 0.500 and 0.333.

Thickness. With $\lambda_t = \lambda_0 t^{-\alpha}$, Eq. (1) gives the thickness ratio between a new ligament and its host as $\lambda_t/\lambda_h = U^{2\alpha}$, again asymptotically and in mean field. Its law does not depend on t, so every generation couples to the frame it attaches to in the same statistical way.

The fill fraction without overlaps, $\rho \simeq \lambda_0 \sum_t \langle\ell_t\rangle t^{-\alpha}$, converges only for $\alpha > 1/2$. This recovers the $\alpha = 1/2$ line of the phase diagram of Ref. [1], but not its second branch: for $\alpha > 1/2$ the structure also jams when $\lambda_0 > [\ell_0\zeta(2\alpha)]^{-1}$ [1]. That branch comes from the depletion of free area, which shortens later ligaments and which the overlap-free estimate ignores. Including depletion in mean field, a ligament nucleated in free area F has length $\ell_t = c(t)(F/t)^{1/2}$, so $dF/dt = -\lambda_0 c(t) F^{1/2} t^{-\alpha-1/2}$ and $\sqrt{F_\infty} = 1 - \frac{\lambda_0}{2}\sum_t c(t) t^{-\alpha-1/2}$, the two-dimensional analogue of the recursion of Ref. [1] and of Eq. (9) below. The structure jams when $\lambda_0 \ge \lambda_0^* = 2/\sum_t c(t) t^{-\alpha-1/2}$. With $c(t)$ measured in zero-thickness runs ($c \to c_\ell = 1.163$), this gives $\lambda_0^* = 0.168$, 0.404, 0.745 and 1.254 for $\alpha = 0.6$, 0.75, 1 and 1.5.

In finite-width simulations, where ligaments nucleate

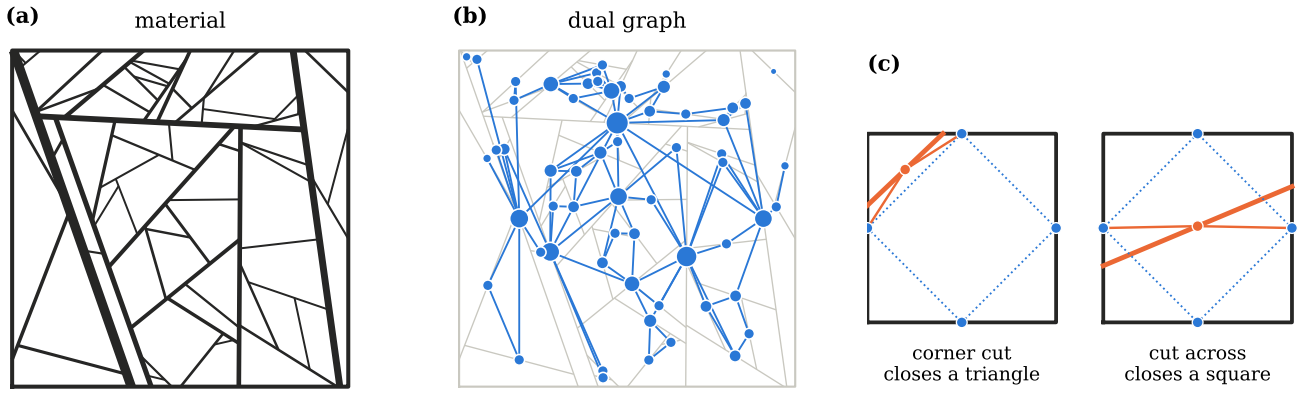


FIG. 1. (a) An SR material with $T = 60$ ligaments. The line width increases with $\lambda_t = \lambda_0 t^{-1/2}$. (b) Its dual graph: each ligament is a node, placed at the ligament midpoint with a size that grows with its degree, and each junction is a link. (c) A new ligament (orange) splits one cell. Dual nodes of the cell's sides are shown in blue, and the dotted lines join sides that meet at a corner. If the new ligament cuts off a corner, it joins two adjacent sides and closes a triangle in the dual. Otherwise it typically joins two sides at distance two and closes a square; in cells with six or more sides the two sides can be farther apart, and 5% of new ligaments close a triangle with a ligament outside the cell.

only in free area and stop at the surface of the ligaments they meet, a run counts as jammed when the free area is exhausted (below 10^{-14}) before step T; λ_0^* is bisected with two realisations per value. At $T = 2 \times 10^4$ we find $\lambda_0^* = 0.232$, 0.363, 0.643 and 0.958. Near $\alpha = 1/2$ the free-area sum converges slowly, its remainder beyond T scaling as $T^{1/2-\alpha}$, and the threshold drifts: at $\alpha = 0.6$ it falls to 0.214 at $T = 10^5$, while at $\alpha = 0.75$ it is stable (0.359). The formula of Ref. [1] gives 0.189, 0.405, 0.643 and 0.879. The depletion recursion is within 11% at $\alpha = 0.75$ but overestimates the threshold by 16–31% at $\alpha \geq 1$, where jamming is decided by the first few thick ligaments. The overlap-free estimate is too low by a factor of about 1.7. Thickness removes area and stops growth in the jammed phase, but it does not change the rule by which cells are split (Sec. VII). In what follows the network is the zero-thickness one unless stated otherwise.

III. A FACE-LOCAL GROWTH RULE: PREFERENTIAL ATTACHMENT OR FITNESS?

In the dual graph, step t reads as follows: choose a face (a cell) with probability equal to its area, choose the two sides of that face that the chord meets, and join the new node t to the two ligaments that carry them [Fig. 1(c)]. The new node never links to a node outside the chosen face. It is a local rule in the sense of Ref. [4], with faces in place of neighbourhoods. Which two sides are hit is decided by the geometry, and two different mechanisms are compatible with the bookkeeping.

Incidences equal degree. Call an incidence a pair (ligament i, cell K) in which i contributes a side to K. A new ligament has two incidences, one on each side, and two links. When a later ligament ends on i, it adds one link to i and cuts one of i's sides in two, adding one incidence. For a ligament whose ends are both interior, the number of incidences therefore equals the degree k_i at all times. Each step creates two hits and four incidences, so the total number of incidences is $4t$, in agreement with $\langle n \rangle = 4$. This identity is exact.

Mechanism A: preferential attachment on incidences. If every incidence were equally likely to be hit, node i would gain links at rate $2k_i/4t$, linear preferential attachment with

$$\frac{dk_i}{dt} = \frac{k_i}{2t}, \qquad k_i(t) = 2\sqrt{t/i}, \tag{2}$$

the Barabási–Albert rule with $m = 2$ [18]. Its master equation [19, 20] gives

$$p_k = \frac{12}{k(k+1)(k+2)}, \qquad P(k' \geq k) = \frac{6}{k(k+1)}, \tag{3}$$

for $k \geq 2$, so $\gamma = 3$ and $K^2 P(k \geq K) \to 6$. Each hit adds an incidence and therefore raises the future hit rate: early hits beget late hits.

Mechanism B: length as fitness. In the geometry, a chord ends on a side with a probability that grows with the side's length, and the total length that ligament i exposes is its length ℓ_i, fixed at birth. Being hit splits a side but adds no length, so there is no reinforcement. Two ends land per step on a total length $L(t) = c_L\sqrt{t}$, so the hit rate per unit length is $2/L(t)$, and the degree is a sum of independent rare events with mean

$$\langle k_i \rangle - 2 = \frac{4\ell_i}{c_L}\left(\sqrt{T} - \sqrt{i}\right). \tag{4}$$

This is the "more opportunities to be hit" argument of Ref. [1], made quantitative. It is a fitness model [8] in which the fitness is the birth length. The cell hierarchy assigns it: $\langle \ell_i \rangle = c_\ell i^{-1/2}$, so that, with $4c_\ell/c_L = 2$,

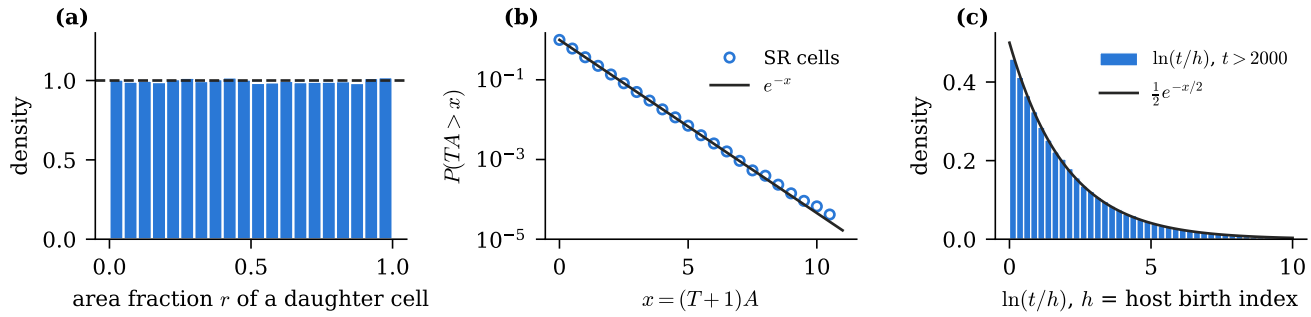


FIG. 2. The cell hierarchy of the SR construction (six realisations, $T = 2 \times 10^4$). (a) The area fraction of a daughter cell is uniform. (b) Rescaled cell areas follow e^{-x}, the spacing law of uniform points. (c) The birth index h of the ligament hit by a new ligament t follows the mean-field law $h/t \to U^2$, for which $\ln(t/h)$ has density $\frac{1}{2}e^{-x/2}$.

Eq. (4) reduces on average to $k_i \simeq 2\sqrt{T/i}$, the same growth law as Eq. (2). The randomness, however, sits in the birth length, $x = \ell_i/\langle \ell_i \rangle$, rather than in the history of hits. With $k \simeq 2x\sqrt{T/i}$ and i uniform,

$$P(k \geq K) \simeq \frac{4\langle x^2 \rangle}{K^2}, \tag{5}$$

again $\gamma = 3$, with an amplitude set, in mean field, by the spread of birth lengths. A hub is predictable at birth from its length.

Which mechanism the geometry realises. Both mechanisms predict $\gamma = 3$ and the same mean growth law, so the exponent cannot decide between them. We apply three tests (three realisations with $T = 10^5$ unless stated). (i) Mechanism B makes $k_i - 2$ a Poisson variable with the mean μ_i of Eq. (4), so the dispersion $D = \langle (k_i - 2 - \mu_i)^2 \rangle / \langle \mu_i \rangle$, taken around each ligament's own prediction, should be one. In bins $\mu_i \in [2,4), [4,8), \ldots, [64,128)$ we measure $D = 0.92$–0.94, 0.93–0.96, 1.01–1.03, 1.07–1.12, 1.05–1.38 and 0.92–1.06, the last two bins holding 300 and 90 ligaments per realisation; no growth of D with μ is resolved. (The variance-to-mean ratio within a bin is larger, 1.05 to 4.7, only because ligaments in a bin have different means.) The mean is also close: the ratio of received to predicted hits is 1.005–1.011 for $\mu_i < 4$ and 1.006–1.034 above, so old ligaments receive about 2% more hits than the mean-field rate $2/L(t)$ gives. Under mechanism A no birth attribute absorbs the variance: in the surrogate of Sec. V, which realises it exactly, the variance at fixed age exceeds the mean by a factor 10 for ligaments born at $i \simeq 1000$ (mean degree 16–17) and 51–91 for those born at $i \simeq 100$ (mean 66–68). (ii) Let e_i be the number of hits on ligament i during $[i, 2i)$ and l_i the number during $[2i, T)$. After subtracting the expectations of Eq. (4), e_i and l_i are uncorrelated, both overall ($r = -0.027$, -0.015 and 0.045) and for the oldest ligaments with $\mu_i > 16$, where the test has most power ($r = -0.017$, 0.032 and 0.094, about 1500 ligaments per realisation). In the surrogate the same correlation at fixed age is 0.46–0.50. (iii) At $T = 10^6$ (five realisations) the measured $K^2 P(k \geq K)$ is 5.78 ± 0.02, 5.75 ± 0.03 and 5.75 ± 0.06 at $K = 32$, 64 and 128. The exact finite-K value of Eq. (3), $6K/(K+1)$, is 5.82, 5.91 and 5.95, so the data lie below preferential attachment by 5σ at $K = 64$. Drawing each degree as 2 plus a Poisson variable with the mean of Eq. (4) and the actual birth lengths gives 5.54–5.57, 5.56–5.62 and 5.59–5.67. Multiplying the mean by the measured ratio of received to predicted hits in each μ bin (1.01–1.025 for $\mu > 2$) gives 5.78, 5.77 and 5.70 (three realisations), which reproduces the data. The tail amplitude is therefore that of the Poisson fitness model with the 2% mean-field correction.

The geometric SR dual therefore grows by mechanism B: a face-local rule in which old nodes are hit in proportion to a fitness fixed at birth. Mechanism A is what the same face-local bookkeeping produces when geometry is removed (Sec. V). In two dimensions the two coincide in the exponent because the hierarchy makes the fitness scale as $i^{-1/2}$, exactly the growth law of preferential attachment with $m = 2$. They differ in their fluctuations, and they differ in three dimensions, where the two mechanisms predict different exponents (Sec. VIII).

Over all ligaments the ratio of measured to predicted degree in Eq. (4) averages 1.006, with correlation 0.95 [Fig. 3(b)]. The degree of the cohort born at i decays as $i^{-\beta}$ with $\beta = 0.50$–0.51.

IV. CORNER CUTS, TRIANGLES AND CLUSTERING

A new node closes a triangle in the dual when the two ligaments it joins are already linked. Locally this happens when the chord cuts off a single corner of its cell, since the two sides meeting at that corner belong to linked ligaments.

Integral-geometry formula. Chords through a uniform point at a uniform angle sample lines with density $w(s)\, ds\, d\theta/(\pi A_K)$. For a fixed direction θ, order the corners by their projection on the normal. The chord cuts

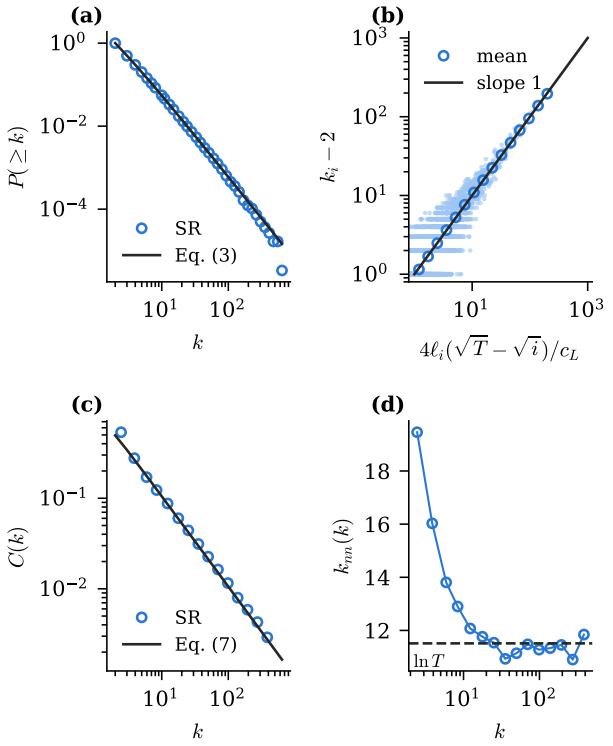


FIG. 3. Dual graph of the SR construction, three realisations with $T = 10^5$. (a) Cumulative degree distribution (circles) and Eq. (3) (line); on this scale it cannot be told apart from Eq. (5). (b) Degree against the prediction (4) from birth length and age (light dots: a random sample of ligaments; circles: binned means; line: slope one). (c) Clustering spectrum (circles) and Eq. (7) with $p_0 = 0.494$, $p = 0.53$ (line). (d) Mean neighbour degree; the dashed line is $\ln T$.

off the lowest corner alone when its offset lies between the projections of the lowest and second-lowest corners. The integral of $w(s)$ over that range is the area $A_{\rm lo}(\theta)$ cut off by the line through the second-lowest corner, and likewise $A_{\rm hi}(\theta)$ at the other end. Hence

$$P_{\rm corner}(K) = \frac{1}{\pi A_K} \int_0^\pi \left[A_{\rm lo}(\theta) + A_{\rm hi}(\theta)\right] d\theta. \qquad (6)$$

The line measure is the standard one of integral geometry [21], weighted here by the chord length because the line passes through a uniform point. For a triangle every chord cuts off a corner, so $P_{\rm corner} = 1$. For a square, Eq. (6) gives $2\ln 2/\pi = 0.4413$ in closed form. Numerical quadrature of Eq. (6) gives 0.303 for a 1×4 rectangle and 0.232 and 0.136 for regular pentagons and hexagons. Evaluated on the simulated SR cells, the mean of Eq. (6) by corner number $n = 3, \ldots, 7$ is 1.000, 0.461, 0.254, 0.152 and 0.10, and its area-weighted average is 0.447–0.449 (numerical). The frequency of corner cuts measured directly in the growth is 0.447–0.449. This agreement checks the formula and the code against each other; it is not an independent prediction, because the cell shapes are taken from the simulation.

Triangle bookkeeping. A corner cut creates a triangular daughter unless the parent was already a triangle, which can only split into $(3, 4)$. The number fraction of triangles is therefore $f_3 = \langle P_{\rm corner}\rangle - w_3$, where w_3 is the area fraction of triangles. Measured values are $0.30 = 0.449 - 0.147$.

Non-local closures. The fraction of new nodes that close a triangle is 0.496, not 1/2. Corner cuts account for 0.448. The other 0.048 join two non-adjacent sides of their cell whose ligaments already meet outside it. The remaining 0.50 of new nodes join two ligaments at distance two or more around the cell: distance two closes a square, and in cells with $n \geq 6$ the two sides can be three steps apart around the face.

Clustering spectrum. Links between existing nodes are never added. A node therefore acquires triangles only at birth, with probability p_0, or when it is hit and the other end of the new ligament lands on one of its neighbours, with probability p per hit. With constant p the number of triangles at node i is $e_i = p_0 + p(k_i - 2)$, and

$$C(k) = \frac{2[p_0 + p(k-2)]}{k(k-1)} \simeq \frac{2p}{k}, \qquad (7)$$

the hierarchical form produced by triadic closure [4, 5, 7]. Counting the per-hit triangle events directly, p is not constant: it is 0.41 when the hit node has degree 2, 0.48 for degrees 3–4, 0.53 for 5–9, and 0.56–0.58 for degrees 10 and above (two realisations, $T = 10^5$, each bin $\geq 10^4$ hits). Hits on hubs are more likely to land next to a neighbour because hubs are long and carry many short stems. With $p_0 = 0.494$ and a single fitted $p = 0.53$ (0.56 when the fit is restricted to $k > 7$), Eq. (7) is within 2% of the data for $k \geq 15$ [Fig. 3(c)]. The mean clustering is 0.38. At $k = 2$ the measured $C = 0.536$ exceeds p_0, because corner-cut ligaments are short and rarely hit again. The degree assortativity is $r \simeq -0.02$. The mean neighbour degree $k_{nn}(k)$ falls from 19.5 at $k = 2$ to a plateau of 11.1–11.5 for hubs, close to the Barabási–Albert value $(m/2)\ln T = \ln T = 11.5$ [Fig. 3(d)].

In the language of local rules, the SR dual mixes two moves. With probability $\simeq 0.45$ the new node attaches to both ends of a link (the corner), as in the edge-attachment model of Dorogovtsev, Mendes and Samukhin [6], except that the link is selected with probability proportional to the area of its cell rather than uniformly. With probability $\simeq 0.50$ it attaches to two nodes at distance two or more on a face cycle, mostly closing a square, and with probability 0.05 it closes a triangle non-locally. The edge-attachment model has $\gamma = 3$ and clustering 0.74 [6, 22]. The SR dual has the same γ and half the clustering; the other half of the closures went into squares. It is also the two-dimensional relative of the random Apollonian network [23, 24], in which faces are selected and a node's degree is proportional to the number of faces it bounds.

V. WHAT TOPOLOGY ALONE REPRODUCES

To separate the role of geometry, we grew a purely topological surrogate. Each face is stored as a cyclic list of the nodes that carry its sides. At each step a face is chosen uniformly. If it is a triangle, two of its sides are joined; otherwise two adjacent sides are joined with probability p_4, or two non-adjacent sides with probability $1-p_4$. The face is then split into two, with $n_1+n_2 = n+4$ as in the tessellation. No lengths or positions are used.

Because every side of the chosen face is equally likely to be hit, every incidence is equally likely to be hit, and the surrogate realises mechanism A of Sec. III exactly. Every face-local variant we tried (faces chosen uniformly, by number of sides, or by a random-split area) reproduces Eq. (3): $\gamma_{\rm MLE}(k \geq 10) = 2.88$–$2.91$ and the same K^2P amplitudes at $K \leq 32$. The single parameter $p_4 = 0.29$ also gives a mean clustering of 0.386 (SR, 0.377) and places $C(k)$ and $k_{nn}(k)$ on the SR curves. The surrogate reaches these statistics through reinforcement, as test (ii) of Sec. III shows, whereas the geometric model reaches them through length. The low-order degree statistics therefore do not identify the mechanism.

The surrogate differs on distances. At $T = 10^5$ its mean distance is 14.1 against 8.69 for SR (Table I). Both grow logarithmically, but with prefactors 1.43 and 0.60 [Fig. 5(a)]. In the geometric model chords hit long sides preferentially, and long sides belong to old, long ligaments, so new nodes attach close to the hubs. In the surrogate, sides are hit regardless of the age of the ligament that carries them, and repeated cuts across faces can produce long chains of squares. Geometry thus matters for the properties that depend on which particular sides are hit, such as distances, and not for $P(k)$, $C(k)$ or $k_{nn}(k)$.

VI. FEWEST-TURN PATHS AND THEIR MULTIPLICITY

A shortest path in the dual graph is a route through the material that changes ligament the fewest times. For a random graph with given degrees, the mean number of shortest paths between two nodes is $\langle\mu\rangle \simeq c_2/e$ with $c_2 = \langle k(k-1)\rangle/\langle k\rangle$, and local growth rules depart from this value in a rule-dependent way [25]. For $\gamma = 3$, c_2 grows as $\ln T$.

Table I lists the measurements for the SR dual, its degree-preserving randomisation (SR*) and the edge-attachment (DM) model [6], each averaged over six realisations and 100 sources per realisation. The randomised graph sits on the configuration-model value, $\langle\mu\rangle/(c_2/e) = 1.00$–$0.90$. The SR dual exceeds it, by a factor that grows from 2.0 ± 0.3 at $T = 10^3$ to 6.4 ± 0.9 at $T = 10^5$, and the DM model stays within 0.87–1.20. Because μ has a heavy tail, $\langle\mu\rangle$ is dominated by rare pairs; the mean logarithm is better determined. The excess $\Delta\langle\ln\mu\rangle = \langle\ln\mu\rangle_{\rm SR} - \langle\ln\mu\rangle_{\rm SR^*}$, taken at the same degree sequence, grows from 0.40 to 1.03 over the same range.

To ask where the excess comes from, we varied the fraction of new nodes that close a triangle at fixed construction and measured $\Delta\langle\ln\mu\rangle$ against each network's own randomisation [$T = 3\times10^4$, six realisations, Fig. 5(b)]. In the topological surrogate, raising the corner-cut probability p_4 from 0.1 to 0.9 raises the triangle-closing fraction from 0.24 to 0.97 and lowers the excess monotonically, from 2.09 ± 0.17 to 0.19 ± 0.03. In the geometric model, the angle set $\{0, \pi/2\}$ closes no triangles and gives an excess of 2.87 ± 0.13; the set $\{0, \pi/3, 2\pi/3\}$ closes 0.23 and gives 2.43 ± 0.32; uniform angles close 0.49 and give 0.83 ± 0.07. Square-closing splits are thus necessary for the redundancy, but their number alone does not set it; their arrangement matters: selecting cells uniformly ($q = 0$) raises the triangle fraction to 0.62 but also raises the excess, to 1.50 ± 0.19. Uniform selection splits small cells as often as large ones and stacks squares locally, which the area weighting of the SR rule avoids. The DM model, which closes only triangles, has no excess.

VII. WHAT MOVES THE DEGREE EXPONENT

Within mechanism B the exponent follows from two scalings: how the fitness of a ligament depends on its birth index, and how its hit rate depends on its fitness. If the hit rate is proportional to ℓ_i^a and $\ell_i \propto i^{-1/2}$, the total rate $\sum_j \ell_j^a$ grows as $t^{1-a/2}$ for $a < 2$, so $k_i \propto \ell_i^a t^{a/2} \propto (t/i)^{a/2}$, giving $\beta = a/2$ and $\gamma = 1 + 1/\beta = 1 + 2/a$. In two dimensions with uniform angles $a = 1$, so $\gamma = 3$. Any rule that keeps the hit rate proportional to length and the birth length proportional to the local cell size leaves γ at 3.

Exponents are estimated with the discrete maximum-likelihood approximation $\hat\gamma = 1 + n[\sum_i \ln(k_i/(k_{\rm min} - 1/2))]^{-1}$ over the n nodes with $k \geq k_{\rm min}$ [26], with standard error $(\hat\gamma-1)/\sqrt{n}$. For a distribution with the curvature of Eq. (3) this estimator is biased low at small $k_{\rm min}$: on 10^5 samples drawn exactly from Eq. (3) it returns 2.77 ± 0.02, 2.87 ± 0.03, 2.91 ± 0.05 and 2.91 ± 0.10 for

TABLE I. Shortest-path statistics of the dual graph: mean distance $\langle d\rangle$, $\langle\mu\rangle/(c_2/e)$ and $\langle\ln\mu\rangle$, mean ± standard error over six realisations (100 sources each). Each entry gives SR / SR*, where SR* is the degree-preserving randomisation of the SR dual; standard errors of the SR* entries are below 0.03.

T	c_2	$\langle d\rangle$	$\langle\mu\rangle/(c_2/e)$	$\langle\ln\mu\rangle$
10^3	7.9	5.78/4.48	$2.0 \pm 0.3/1.00$	$1.09 \pm 0.06/0.70$
3×10^3	9.3	6.60/4.98	$2.9 \pm 0.5/1.00$	$1.39 \pm 0.07/0.80$
10^4	11.0	7.18/5.31	$3.3 \pm 0.4/0.94$	$1.55 \pm 0.06/0.87$
3×10^4	12.3	7.98/5.65	$5.4 \pm 1.0/0.93$	$1.77 \pm 0.07/0.94$
10^5	14.1	8.69/6.06	$6.4 \pm 0.9/0.90$	$2.03 \pm 0.06/1.00$

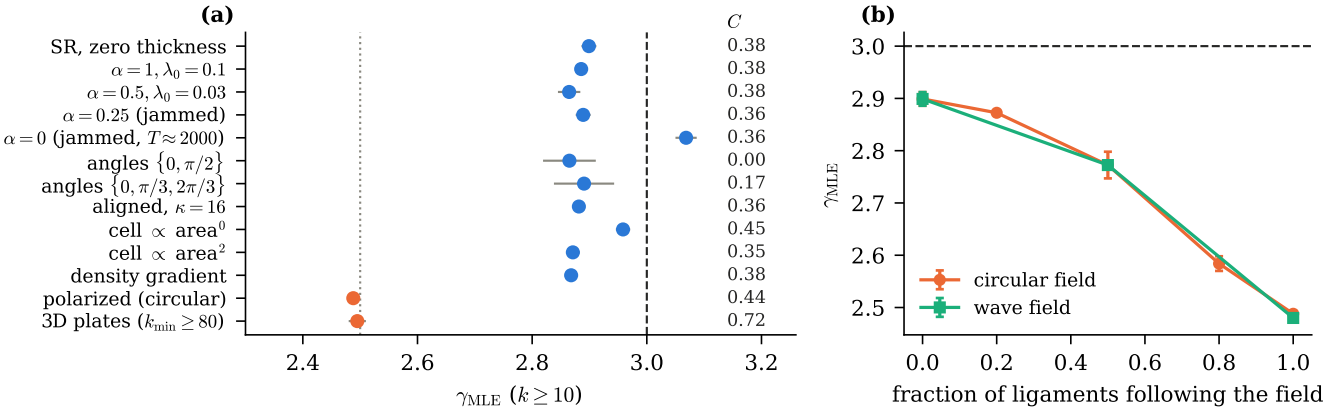


FIG. 4. (a) Maximum-likelihood exponent ($k \geq 10$) for the design rules of Ref. [1], $T = 3 \times 10^4$, two realisations (the jammed $\alpha = 0$ case stops at $T \simeq 2000$; 3D plates: $T = 2 \times 10^6$, three realisations, $k_{\min} = 80$–640). The number on the right is the mean clustering coefficient. Error bars: half the difference between the two realisations (3D: range over $k_{\min}$). Dashed line: $\gamma = 3$; dotted line: $\gamma = 2.5$. The uniform SR control gives 2.87–2.91 at this size, the downward bias of the estimator discussed in Sec. VII. (b) Exponent against the fraction of ligaments whose angle follows a circular or a wave orientation field ($T = 3 \times 10^4$, two realisations; error bars: half their difference); the remaining ligaments have uniform angles.

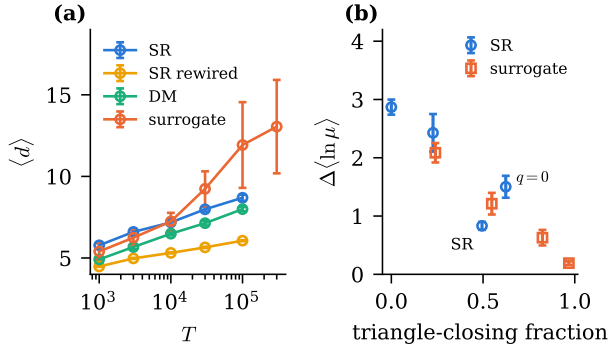


FIG. 5. (a) Mean distance in the dual graph against T for the SR construction, its degree-preserving randomisation, the edge-attachment (DM) model [6] (six realisations) and the topological face surrogate with $p_4 = 0.29$ (two to three realisations). (b) Excess of the mean log-multiplicity over each network's degree-preserving randomisation against the fraction of new nodes that close a triangle, $T = 3{\times}10^4$, six realisations: SR rule variants (circles) (angle sets $\{0, \pi/2\}$, $\{0, \pi/3, 2\pi/3\}$, uniform angles, and uniform cell selection $q = 0$) and the face surrogate with $p_4 = 0.1$, 0.29, 0.6, 0.9 (squares). Error bars: standard errors.

$k_{\min} = 6$, 10, 20 and 40. The SR model at $T = 10^5$ gives 2.77–2.79, 2.86–2.90, 2.94–2.97 and 2.90–3.03. Statements that $\gamma = 3$ therefore rest on comparison with these controls, not on the raw value of $\hat{\gamma}$.

We tested the design rules of Ref. [1] at $T = 3 \times 10^4$, two realisations each [Fig. 4(a)]:

- Thickness in the tunable-density phase ($\alpha = 0.5$–1, $\lambda_0 = 0.01$–0.1, $\rho = 0.12$–0.45) gives $\gamma_{\mathrm{MLE}} = 2.85$–2.92, $\beta = 0.50$–0.52, $C = 0.376$–0.378, and a triangle closure of 0.49–0.50.
- In the jammed phase ($\alpha = 0$–0.4), growth stops when the free area is exhausted (at $t \simeq 2000$ for $\alpha = 0$, $\lambda_0 = 0.02$). Jamming truncates T; it does not change γ (2.89–2.94 wherever the statistics allow) or C (0.35–0.37).
- Angle sets change clustering but not γ. With angles $\{0, \pi/2\}$ all cells are rectangles, so $C = 0$ exactly and $\gamma = 2.87$; this is the rectangular fragmentation of Ref. [17]. With $\{0, \pi/3, 2\pi/3\}$, $C = 0.17$.
- Global alignment (von Mises, $\kappa \leq 16$), cell selection proportional to area^q ($q = 0, 0.5, 2$), and density gradients give $\gamma = 2.86$–2.96; $q = 0$ and $q = 2$ give $C = 0.45$ and 0.35.

The angle set and the selection weight tune clustering between 0 and 0.45; the exponent stays at 3.

Coherent orientation fields. The polarized rule of Ref. [1], in which the angle follows a prescribed field $\theta(x, y)$, breaks the proportionality between hit rate and length. Locally parallel newcomers cannot hit each other; they end on the ligaments that are misaligned with the local field, and for straight ligaments in a curved field the misalignment grows with length. The degree then grows faster than length, and birth lengths decay more slowly than $i^{-1/2}$: $\langle \ell_i \rangle \propto i^{-0.35}$ for the circular field and $i^{-0.39}$ for the shortest wave, against $i^{-0.50}$ with uniform angles. Fitting $\ln(k-2)$ against $\ln \ell$ for the oldest 5% of ligaments with $k > 4$ gives slopes $a = 1.43$–1.63, against 1.03–1.05 for uniform angles. With $a \simeq 1.5$ the argument above with $\ell_i \propto i^{-1/2}$ predicts $\beta \simeq 0.75$ and $\gamma \simeq 2.3$. With the measured birth-length decay $i^{-\delta}$ the same argument gives $\beta = a\delta \simeq 0.53$–0.6 and $\gamma \simeq 2.7$–2.9, now on

the other side of the measured values. Under fields the length at fixed birth time is broadly distributed, and the exponent is set by the joint distribution of length and age rather than by the two mean scalings; we have no theory for it. For a circular field and for waves $\theta = \pi(x+y)/w$ with $w = 0.125$–1 we find $\gamma \simeq 2.44$–2.51 and cohort exponents $\beta \simeq 0.6$–0.67, stable from $T = 10^4$ to 2×10^5. At $T = 10^6$ the maximum-likelihood exponent is 2.43–2.50 ($k \geq 10$ to $k \geq 40$) for the circular field and 2.36–2.47 for the shortest wave ($w = 0.125$), while the uniform control gives 2.88–2.98, so the effect does not wash out at this size. The exponent decreases continuously as the fraction of ligaments following the field increases [Fig. 4(b)].

VIII. THREE DIMENSIONS

With plates, the SR construction is the iterated division of convex polyhedra. A cell is selected with probability equal to its volume, and it is cut by the plane through a uniform point with an isotropic normal. The argument of Sec. II carries over unchanged: for a fixed normal, the offset of the plane has density $A(s)/V_K$, with $A(s)$ the section area, so the volume fraction of a daughter is uniform. The cell hierarchy is again a binary search tree, with exponential volumes and split depth $2\ln t$. We measure a daughter-fraction variance of 0.0833 (1/12) and a depth slope of 2.01. A plane that crosses m faces of a simple polyhedron changes (F, V, E) by $(m+2, 2m, 3m)$ summed over the two daughters. Hence the mean numbers of faces, vertices and edges per cell are $2+\bar m$, $2\bar m$ and $3\bar m$, the counterpart of $\langle n\rangle = 4$ (measured 6.585, 9.170, 13.755). Unlike in two dimensions, the number m of plates met by a new plate is not fixed by topology. Its mean converges to $\bar m = 4.577$ for plates away from the walls. Plate areas decay as $a_t \simeq c_a t^{-2/3}$ with $c_a = 1.23$, so the total area grows as $3c_a t^{1/3}$ and the area distribution is $P(a) \sim a^{-5/2}$ (maximum-likelihood exponent 2.49–2.51). The first plate has $\langle a_1\rangle = 0.941$.

In three dimensions mechanisms A and B of Sec. III predict different exponents. Counting faces, a new plate creates $2+\bar m$ incidences and hands out $\bar m$ links. Mechanism A, a uniform hit rate per incidence, would give $b = \bar m/(2+\bar m)$ and $\gamma = 2 + 2/\bar m = 2.44$. Under mechanism B the hit rate is set by the plate area, which is fixed at birth. With a rate per unit area $\bar m/2A(t)$ on each side,

$$k_i = o_i + \frac{\bar m}{2}\, x_i \Big[\Big(\frac{T}{i}\Big)^{2/3} - 1\Big], \qquad \gamma = \frac{5}{2}, \tag{8}$$

where o_i is the out-degree at birth and $x_i = a_i/\langle a_i\rangle$. This is the fitness argument of Sec. VII with area in place of length; its general form is $\gamma = (2d-1)/(d-1)$, the exponent of random Apollonian networks in d dimensions [27]. The two mechanisms agree in two dimensions only because $m/(2+m) = (d-1)/d$ when $m = 2$. Geometry decides between them: an old plate is tiled by faces of linear size $t^{-1/3}$, so its number of faces grows as $t^{2/3}$, whereas mechanism A requires the faster growth $t^{\bar m/(2+\bar m)} = t^{0.694}$. Equivalently, the hit rate per face is not uniform: a newborn plate has one large face per side and receives more hits per face than an old, finely tiled plate. Per unit area the rate is nearly uniform, 1.04 for old plates and 0.93–0.95 for young ones in units of the mean, which is mechanism B with a few-per-cent correction of the same sign as in two dimensions. We find (three realisations, $T = 2\times10^6$):

- the faces of a fixed old plate grow as $t^{0.663\text{–}0.668}$ (range over three realisations and three birth cohorts, $i \in [10, 10^4)$), against 0.694 for mechanism A; a reinforcement surrogate with the same out-degrees gives 0.688–0.690 and a fitness surrogate 0.658–0.661;
- Eq. (8) predicts the hits received by each plate with ratio 1.015 and correlation 0.99;
- at fixed age the hits scale with birth area with elasticity 0.94–0.95;
- hits in $(i, 2i]$ predict hits in $(2i, T]$ at fixed area only weakly (log–log slope 0.025–0.030), against 0.55 in a reinforcement surrogate with the same out-degrees.

The dispersion of the received hits around each plate's own prediction from Eq. (8) is 1.5–4 times Poisson, rising with the predicted mean, against 100–380 times for the reinforcement surrogate. It corresponds to a weak per-plate rate factor with a coefficient of variation of 0.16 to 0.07, which we attribute to the local cell-size environment. The degree of a plate is therefore a fitness (its area) multiplied by an age factor. The degree distribution follows the fitness surrogate within 3–7% up to $k \approx 800$, while it departs from the reinforcement surrogate by up to 23%. The maximum-likelihood exponent of all plates is 2.48–2.51 for $k_{\min} = 80$–640, with standard errors per realisation of 0.014, 0.023, 0.038 and 0.064 at $k_{\min} = 80$, 160, 320 and 640. The fitness and reinforcement surrogates, whose asymptotes are 2.5 and 2.44, read 2.48–2.53 and 2.40–2.46 at the same cuts. The exponents alone separate the mechanisms only at $k_{\min} \leq 160$, by two to four standard errors; the shape of the distribution separates them more clearly, with ratios to the fitness surrogate of 0.97–1.07 and to the reinforcement surrogate drifting from 1.06 to 0.79 over $K = 10$–2000. Restricting the fit to plates away from the walls biases it upward, to 2.67–2.72, 2.70–2.83, 2.79–3.00 and 2.96–3.18 at the same cuts: the oldest, largest plates touch the walls (only 18% of plates born before $i = 100$ are interior), and they are hit by area like the others. Three-dimensional SR materials therefore have $\gamma = 5/2$. The host of a new plate satisfies $h/t \to U^3$ (measured $P(h/t \leq 0.01) = 0.217$, against 0.215), so the thickness ratio to the host is $U^{3\alpha}$.

A plate nucleated in free volume F has area $a_t = c(t)(F/t)^{2/3}$, with $c(t) \to c_a = 1.23$ measured in zero-

thickness runs, so the free volume obeys

$$\frac{dF}{dt} = -\lambda_0 c(t) F^{2/3} t^{-\alpha-2/3},$$
$$F_\infty^{1/3} = 1 - \frac{\lambda_0}{3}\sum_t c(t) t^{-\alpha-2/3}. \tag{9}$$

It matches the simulated free volume to within 0.4% in the tunable phase. The sum converges only for $\alpha > 1/3$, which places the jamming boundary at $\alpha = 1/3$ in three dimensions, and for $\alpha > 1/3$ Eq. (9) predicts a second branch, $\lambda_0^* \simeq 3/[c_a\zeta(\alpha + 2/3)]$, the analogue of the λ_0 branch in two dimensions; we have not located it in simulations. For small λ_0 in the tunable phase, $\rho \simeq \lambda_0 c_a \zeta(\alpha + 2/3)$.

Clustering is high because a new plate's section polygon is a cycle of plates already in contact. Each newcomer closes 4.39 triangles along its polygon and 2.52 more through contacts outside its cell. Each received link brings $q = 3.26$ triangles. Then $C(k) = 2[e_0 + q(k - o)]/[k(k-1)] \to 6.5/k$, which agrees with the data within 1% for $k \geq 16$, and $\langle C\rangle = 0.72$. The three-dimensional random Apollonian network has the same γ, $C(k) = 6/k$ and $\langle C\rangle = 0.82$. The mean neighbour degree decreases with k and grows as $T^{1/3}$.

Thickness and jamming change neither γ (2.49–2.55) nor C (0.71–0.72). With three orthogonal orientations every cell is a box and every plate a rectangle ($m = 4$). The two counting arguments then coincide at $b = 2/3$, and $\gamma \simeq 2.5$. These boxes cluster ($C = 0.55$), whereas rectangles in two dimensions do not. Beams, which need a capture rule, are discussed among the open problems in Sec. X.

IX. CONSEQUENCES FOR STRUCTURE–PROPERTY RELATIONS

At fixed α, the thickness of ligament i is $\lambda_0 i^{-\alpha}$ exactly, and its degree and length both scale as $i^{-1/2}$ on average [Eq. (4)]. The marginal laws reported in Ref. [1], an energy that decays as $i^{-(3/4+\alpha/2)}$ with the introduction time and the corresponding powers of k, ℓ and λ, are one age law written in different variables. A product $U \sim k^a\lambda^b$ reproduces the age law for every α only if $a/2 + \alpha b = 3/4 + \alpha/2$, that is $a = 3/2$ and $b = 1/2$, the reported exponents. Because $k \propto \ell$ at fixed age [Eq. (4)], the same data are described equally well by $U \sim \ell^{3/2}\lambda^{1/2} f(\theta)$. Degree, length and age cannot be separated from cohort scaling alone.

Separating them requires the per-ligament energies themselves: a within-cohort regression of U on k, ℓ and λ would show whether degree adds anything beyond length and age. In a slender-beam frame model that we used as a check (Euler–Bernoulli beams, rigid T-junctions, $T \leq 5000$), fits on degree and on length were indistinguishable ($R^2 = 0.634$ and 0.631); a frame of slender beams is not the continuum finite-element model of Ref. [1], so we do not report its exponents.

The same reasoning applies to design. Thickness is a function of birth order, and degree and length are statistically slaved to birth order, so α and λ_0 act on the material through the age of each ligament on a fixed network. Connectivity becomes a design variable only when the rule is changed so that the hit rate stops being proportional to length, as with coherent orientation fields, or when the dimension changes the relation between fitness and birth order (Secs. VII and VIII).

X. CONCLUSIONS

The scale-rich construction joins two known structures. Geometrically it is Cowan's recursive division of cells [9]; as a graph it grows by a face-local rule in which each new node joins two nodes on the boundary of one face. The recursion explains the cell hierarchy: uniform split fractions, a binary-search-tree genealogy and exponential areas. It also assigns each ligament its length at birth, $\ell_i \propto i^{-1/2}$, and that length acts as a fitness: the degree is a Poisson count of hits with mean proportional to length times remaining time, without reinforcement. The k^{-3} tail, which preferential attachment on face incidences would also give, follows from that fitness, and the two mechanisms are separated by the fluctuations of the degree rather than by its exponent. The local rule explains the rest of the network: a hierarchical clustering spectrum from corner cuts and an excess of shortest paths from square-closing splits. Keeping both references in view settles which features are fixed by the class and which can be designed. The exponent $\gamma = 3$ (and 5/2 for plates in three dimensions) and the equality $\langle n\rangle = 4$ are fixed, and in mean field so are the host-age law $h/t \to U^2$ and the stationarity of the thickness ratio. Clustering is designable through the angle set and the selection weight. The degree exponent is designable only through rules that make the hit rate grow faster than length.

Several points remain open. The rate $2/L(t)$ per unit length is a mean-field rate: old ligaments receive about 2% more hits than it predicts, in two and in three dimensions, which sets the tail amplitude; we have no theory of this correction. The per-hit triangle probability, which rises from 0.41 to 0.57 with the degree, and the non-local closure fraction 0.048 require the joint law of cell shapes next to long ligaments. Under coherent orientation fields we observe $\gamma \simeq 2.4$–2.5 up to $T = 10^6$; the fitness argument gives the direction of the change but not its value. In three dimensions the mean number of plates met by a new plate, $\bar{m} = 4.577$, has no closed form. For beams, zero-width lines do not meet in three dimensions and a capture rule is needed. If a rod of radius $R_t = \lambda_t/2$ grows until it touches another, it stops after a mean free path $\ell_t \sim 1/S(t)$, where $S(t) = \sum_j \ell_j R_j$ is the total capture cross-section. With $\ell_t \sim t^{-\beta'}$, $S \sim t^{1-\beta'-\alpha}$, and self-consistency gives $\beta' = (1 - \alpha)/2$. A rod is then hit in proportion to

$\ell_i R_i \propto i^{-(1+\alpha)/2}$, so its degree grows as $(t/i)^{(1+\alpha)/2}$ and, as a conjecture, $\gamma = 1 + 2/(1+\alpha) = (3+\alpha)/(1+\alpha)$: for beams the thickness exponent would tune γ. Our simulations up to $T = 10^5$ find hits proportional to rod length but are dominated by the walls, and the capture rule of Ref. [1] is not specified in the main text. Finally, the growth of the shortest-path excess with T is consistent with both a log-quadratic and a weak power law over the range studied.

More broadly, the SR dual shows that a spatial growth process can reproduce the degree statistics of a non-spatial growth class by a different route. Here the area-weighted point and the convexity of the cells deliver each new node to a face, as a random walk would, but the choice of sides within the face is set by length, a property fixed at birth, rather than by accumulated links.

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